

Hasse-Anchored Curriculum for Pairwise Preference Learning

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Abstract

Pairwise preference learning depends not only on how many comparisons are used, but on when different comparison types are introduced. We study this question as a curriculum design problem over pair distributions in a controlled contextual preference benchmark. The Hasse skeleton of a prompt-wise transitive preference relation is the minimal set of cover comparisons that preserves the ordering structure over all comparable pairs in the transitive closure. This makes Hasse pairs a natural late-stage local-ordering anchor, rather than a complete replacement for broad coverage. Empirically, Hasse pairs are hard local constraints and work best after the model has already learned coarse separation and broad coverage. Across matched MovieLens experiments, the strongest practical form among the tested curricula is $\text{Ext} \rightarrow \text{Rnd} \rightarrow \text{Hasse} + \text{Aux}$: extreme large-gap pairs provide rank-distance-easy early supervision, random pairs preserve coverage, and the final Hasse-anchored stage improves late local and top-ranking structure while auxiliary pairs supply complementary non-Hasse supervision. Auxiliary ablations suggest that premature model-dependent hard mining can narrow the effective training distribution and hurt held-out ranking. These results support a stage-wise distribution-shaping view of controlled pairwise preference learning.

1 Introduction

Pairwise preference learning builds on a long line of comparison-based preference and ranking models (Thurstone, 1927; Zermelo, 1929; Bradley & Terry, 1952; Luce, 1959; Plackett, 1975). It is often formulated as a Bradley–Terry-style pairwise ordering problem: given two responses to the same prompt, the model is trained to assign a higher score to the preferred response. When the preference data provide graded or transitive structure, pair difficulty can be organized by preference gap. Large-gap comparisons provide coarse and relatively easy supervision, while local small-gap comparisons require finer discrimination. This makes curriculum learning a natural lens (Bengio et al., 2009; Kumar et al., 2010; Jiang et al., 2015; Graves et al., 2017): one can introduce broad or easy comparisons early and reserve harder local comparisons for later stages.

Pairwise preference learning should not be viewed only as a problem of sampling more pairwise comparisons. In curriculum training, later stages often emphasize selected subsets of pairs, especially local or model-dependent hard pairs (Kumar et al., 2010; Jiang et al., 2015; Graves et al., 2017; Shrivastava et al., 2016; Lin et al., 2017). Such selection can be useful, but it also reshapes the effective training distribution. If the final stage concentrates too narrowly on a biased subset, the model may lose broad order information even while improving on the selected pairs.

This paper studies whether the Hasse skeleton of a prompt-wise preference order can serve as a structural anchor for such late-stage distribution shifts. The Hasse skeleton consists of local cover comparisons: it is not a broad replacement for the full pair distribution, and by itself it can be a hard small-gap training set. Its value is instead that it preserves the local transitive structure needed to keep late-stage refinement tied to the original preference order. We instantiate this idea through the curriculum family

$$\text{Ext} \rightarrow \text{Rnd} \rightarrow \text{Hasse} + \text{Aux}.$$

Throughout the paper, Ext, Rnd, Hasse, and Aux denote both pair selectors and the training stages that use the selected pairs. Ext pairs are large-gap comparisons that serve as rank-distance-easy early supervision for coarse ranking separation: by construction, they compare responses separated by a large preference gap within the same prompt. Rnd pairs preserve broad coverage of the original pair distribution. Hasse pairs are local cover relations of the preference partially ordered set (poset); in our tiered construction, these cover relations correspond to adjacent preference levels and therefore to local, typically small-gap comparisons, but they carry the transitive-ranking anchor needed for late refinement. Aux denotes non-Hasse auxiliary supervision paired with the final Hasse anchor. The key point is that late-stage selected auxiliary data are not necessarily safe by themselves: because they are biased subsets of the non-Hasse pool, they can narrow the effective training distribution and reduce held-out generalization. We therefore use Hasse not merely as another hard subset, but as a structural anchor around which biased auxiliary supervision can be added.

The Hasse skeleton is theoretically special under finite, noiseless, prompt-wise transitive preferences. It contains only the cover comparisons of the preference poset. It is minimal in the sense that correctness on these cover comparisons suffices to recover correctness over all comparable pairs in the transitive closure, while omitting a cover edge can destroy such a prompt-wise guarantee. We therefore use Hasse pairs as a structural anchor: they are the local constraints that should be present when late-stage training refines the induced comparable-pair order.

The experiments show that this anchor is useful in the intended curriculum role. Hasse training from scratch is much worse than random training, confirming that Hasse is hard local supervision rather than a standalone data replacement. After broad warm-up, however, Hasse continuation improves over budget-matched random continuation. We then show that Aux-only late continuation can be unstable, while Hasse+Aux consistently improves over Rnd+Aux when the auxiliary selector is held fixed. The strongest empirical pattern is therefore staged: learn coarse separation with Ext, maintain coverage with Rnd, and use a Hasse-anchored final stage to refine local preference structure without letting selected auxiliary data dominate the late training distribution.

Our contributions are:

- We formulate the Hasse skeleton as a minimal transitive-ranking anchor for prompt-wise transitive preferences and prove that Hasse-containing losses control error over all comparable pairs in the transitive closure.
- We propose a Hasse-anchored curriculum that introduces easy Ext pairs for coarse separation, Rnd pairs for coverage, and a Hasse+Aux final stage that uses Hasse as a structural anchor for selected non-Hasse auxiliary supervision.
- We analyze matched auxiliary selectors for the final Hasse-anchored stage: random coverage pairs (Rnd), extreme easy pairs (Ext), uncertain pairs (Unc), and misranked pairs (Mis).
- In a controlled contextual preference benchmark, we show empirically that Hasse is harmful from scratch but useful after warm-up, that Aux-only late continuation can distort generalization, and that Hasse provides a stronger final-stage anchor than budget-matched random data across several auxiliary selectors.

2 Related Work

Reward modeling from pairwise preferences is one important application of pairwise preference learning. It is commonly trained with Bradley–Terry-style comparison losses (Bradley & Terry, 1952) and is a central component of learning from human feedback (Christiano et al., 2017; Ziegler et al., 2019; Stiennon et al., 2020; Ouyang et al., 2022; Bai et al., 2022; Touvron et al., 2023). Direct preference optimization also uses paired preference data to optimize language models without an explicit learned reward model (Rafailov et al., 2023). Recent work has emphasized that preference data are not uniformly reliable or equally useful throughout training (Singhal et al., 2024). We do not claim downstream reward-model improvements here; instead, our work focuses on the controlled preference learning question: given a controlled transitive preference structure, which pair distributions should be introduced at different training stages?

The curriculum perspective is related to curriculum learning (Bengio et al., 2009; Kumar et al., 2010; Jiang et al., 2015; Graves et al., 2017), where easier examples can be introduced before harder ones or selected adaptively. It is also related to hard-example mining, which deliberately concentrates training on examples that the current model finds difficult (Shrivastava et al., 2016; Lin et al., 2017). In our setting, Unc and Mis play the role of model-dependent hard selectors. The experiments show that these selectors are not automatically beneficial: after a short warm-up, concentrating on hard non-Hasse pairs can hurt held-out ranking relative to coverage-preserving or large-gap auxiliary data.

Our dataset control is based on the transitive structure of preference data. The Hasse dataset is the edge set of the transitive reduction of a preference poset (Davey & Priestley, 2002), closely related to the classical notion of transitive reduction in directed graphs (Aho et al., 1972). Preference and ranking methods have long studied how to learn from pairwise comparison graphs (Freund et al., 2003; Pahikkala et al., 2009; Ailon, 2012), and learning-to-rank methods similarly optimize ranking objectives from pairwise or listwise signals (Joachims, 2002; Burges et al., 2005; Cao et al., 2007; Liu, 2009; Burges, 2010). Rank aggregation and rank-breaking methods study how structured rankings or partial rankings can be represented through pairwise comparisons (Rajkumar & Agarwal, 2014; Chen & Suh, 2015; Shah et al., 2016; Negahban et al., 2016). Our sampling schedule compares the Hasse skeleton with stage-dependent selectors such as extreme, random, uncertain, and misranked pairs, separating the transitive-control role of the Hasse reduction from the empirical roles of coarse separation, coverage, and hard mining.

3 Definition and Notation

3.1 Pairwise Preference Setup

Definition 3.1 (Ranking scores and signed margin). *Let $D_\theta : X \times Y \rightarrow \mathbb{R}$ be a scoring model. For a prompt x and response y_i , write*

$$D_i(\theta) := D_\theta(x, y_i).$$

For a comparable pair (x, y_i, y_j) , define the observed preference sign by

$$s_{ij} := \begin{cases} 1, & \text{if } y_i \text{ is preferred to } y_j, \\ -1, & \text{if } y_j \text{ is preferred to } y_i. \end{cases}$$

Define the signed ranking margin by

$$\Delta_{ij}(\theta) := s_{ij} (D_i(\theta) - D_j(\theta)).$$

The pair is correctly ranked when $\Delta_{ij}(\theta) > 0$ and incorrectly ranked when $\Delta_{ij}(\theta) < 0$. Ties, i.e. $\Delta_{ij}(\theta) = 0$, are treated as boundary cases. When θ is clear from context, we write D_i and Δ_{ij} for $D_i(\theta)$ and $\Delta_{ij}(\theta)$.

Definition 3.2 (Pairwise loss and ranking error). *Let $\mathcal{D}$ be a pairwise preference dataset, i.e., an empirical collection of comparable triples (x, y_i, y_j) together with their observed preference signs s_{ij} . Expectations over $(x, y_i, y_j) \sim \mathcal{D}$ denote empirical averages over this collection. The logistic function is denoted by*

$$\sigma(z) := \frac{1}{1 + \exp(-z)}.$$

The pairwise negative log-likelihood loss is

$$\mathcal{L}_{ij}(\theta) := -\log \sigma(\Delta_{ij}(\theta)).$$

The pairwise preference loss is defined as

$$\mathcal{L}_{\text{rank}}(\theta) := \mathbb{E}_{(x, y_i, y_j) \sim \mathcal{D}} [\mathcal{L}_{ij}(\theta)],$$

and the ranking error is defined as

$$\mathcal{R}(\theta) := \mathbb{E}_{(x, y_i, y_j) \sim \mathcal{D}} [\mathbf{1}[\Delta_{ij}(\theta) < 0]].$$

Proposition 3.1 (Convergence of ranking error). *For any θ ,*

$$\mathcal{R}(\theta) \leq \frac{1}{\log 2} \mathcal{L}_{\text{rank}}(\theta).$$

Consequently, if $\mathcal{L}_{\text{rank}}(\theta_t) \rightarrow 0$ for a sequence $\{\theta_t\}_{t=1}^\infty$, then $\mathcal{R}(\theta_t) \rightarrow 0$.

Proof. The proof is given in Appendix A. □

4 Hasse-Anchored Curriculum Framework

4.1 Method Overview

We use a Hasse-anchored curriculum to separate the roles of easy separation, distributional coverage, and local transitive refinement. For each prompt, we first extract the Hasse dataset from the transitive preference relation. Hasse pairs provide the minimal constraint skeleton whose correct ranking controls the comparable-pair order induced by the transitive relation.

The main curriculum uses three roles. The model first trains on Ext pairs, which are large-gap and rank-distance-easy. It then trains on Rnd pairs, which preserve broad coverage of the pair distribution. Finally, it trains on a Hasse-anchored stage: Hasse pairs provide local cover relations, while auxiliary non-Hasse pairs can add coverage, easy separation, or targeted refinement. Thus, the complete curriculum is Ext→Rnd→Hasse+Aux, while the Hasse-only final stage is the base case used to isolate the late-anchor effect. The main empirical question is whether a staged curriculum can use each pair type at the point where it is most useful.

The final stage is deliberately Hasse-anchored rather than Aux-only. Auxiliary selectors are biased by construction: they keep random coverage pairs, extreme large-gap pairs, boundary cases, or model-dependent errors. Such selected non-Hasse subsets can be useful, but if they are used alone they can shift the late training distribution away from the broader prompt-wise order. The Hasse component is intended to prevent this shift from becoming unanchored by keeping the local cover skeleton present during late refinement. We therefore evaluate Ext→Rnd→Hasse to establish the late-anchor ordering, and then evaluate Ext→Rnd→Hasse+Aux as the full Hasse-anchored curriculum.

4.2 Hasse Data Set

In real-valued score-based ranking, preference data are ultimately embedded into the usual order on $\mathbb{R}$ through the scalar scores assigned to candidate responses. A consistent preference relation must therefore be compatible with this transitive order: if one response is preferred to a second and the second to a third, the first should also be preferred to the third. We make this transitivity assumption explicit before defining the pair subsets used throughout the method.

Definition 4.1 (Transitive preference dataset). *For each prompt x , let Y_x be the finite set of candidate responses equipped with a strict preference relation $\succ_x$. We say that $(Y_x, \succ_x)$ is transitive if, for any responses $y_i, y_j, y_k \in Y_x$,*

$$y_i \succ_x y_j \text{ and } y_j \succ_x y_k \implies y_i \succ_x y_k.$$

Equivalently, $(Y_x, \succ_x)$ forms a strict partially ordered set of responses for the fixed prompt x . For an induced collection of prompt-wise full comparable-pair datasets, we say that $\mathcal{D}_{\text{full}}$ is transitive if $\mathcal{D}_{\text{full}}^x$ is induced by such a transitive relation for every prompt x .

Unless otherwise stated, all preference datasets below are assumed to be finite and prompt-wise transitive in the sense of Definition 4.1. We first define the full comparable-pair dataset and then define the cover relations that form the Hasse dataset.

Definition 4.2 (Full and Hasse datasets). *Given a transitive prompt-wise response set $(Y_x, \succ_x)$, the full pairwise datasets are defined as*

$$\mathcal{D}_{\text{full}}^x := \{(x, y_i, y_j) : y_i \succ_x y_j\}, \quad \mathcal{D}_{\text{full}} := \bigsqcup_{x \in X} \mathcal{D}_{\text{full}}^x.$$

We define the cover relation $\succ_x$ by

$$y_i \succ_x y_j \iff y_i \succ_x y_j \text{ and there is no } y_k \in Y_x \text{ such that } y_i \succ_x y_k \succ_x y_j.$$

The Hasse pairwise datasets are defined as

$$\mathcal{D}_{\text{Hasse}}^x := \{(x, y_i, y_j) : y_i \succ_x y_j\}, \quad \mathcal{D}_{\text{Hasse}} := \bigsqcup_{x \in X} \mathcal{D}_{\text{Hasse}}^x.$$

For a scoring model θ , the error over all comparable pairs and the Hasse ranking error for prompt x are

$$\mathcal{R}_{\text{full}}^x(\theta) := \frac{1}{|\mathcal{D}_{\text{full}}^x|} \sum_{(x, y_i, y_j) \in \mathcal{D}_{\text{full}}^x} \mathbf{1}[D_i - D_j < 0], \quad \mathcal{R}_{\text{Hasse}}^x(\theta) := \frac{1}{|\mathcal{D}_{\text{Hasse}}^x|} \sum_{(x, y_i, y_j) \in \mathcal{D}_{\text{Hasse}}^x} \mathbf{1}[D_i - D_j < 0].$$

Remark. Here “full” refers only to the finite set of comparable pairs induced by the prompt-wise preference relation. The propositions below are therefore within-prompt transitive-control statements; they do not by themselves imply generalization to unseen prompts or to noisy, cyclic preference data. Because the global dataset is the finite disjoint union of prompt-wise datasets, convergence of the prompt-wise ranking errors implies convergence of the corresponding empirical ranking error over their union. We therefore state the structural results prompt-wise and use the induced global dataset without further distinction.

For each prompt x , the partial order can be viewed as a directed graph whose vertices are the candidate responses and whose edges point from a preferred response to a less preferred one. Because the relation is transitive and antisymmetric, this graph is acyclic. Under this graph view, $\mathcal{D}_{\text{full}}^x$ contains all comparable directed edges, i.e., the transitive closure of the prompt-wise preference graph, while $\mathcal{D}_{\text{Hasse}}^x$ keeps only the transitive reduction. The Hasse diagram keeps the same response vertices but removes edges whose preference relation is already implied by a directed path. Thus, it is a minimal reachability-preserving skeleton.

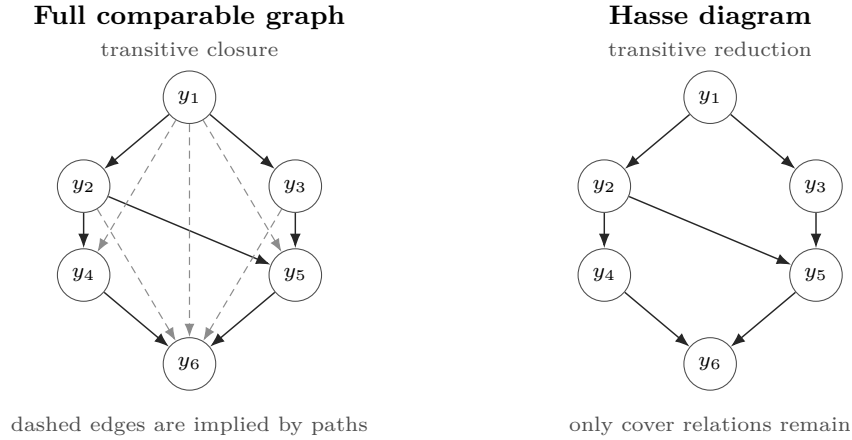


Figure 1: Graph interpretation of the Hasse dataset. A prompt-wise partial order induces a directed acyclic graph over responses. The full comparable graph contains all implied comparable edges, whereas the Hasse diagram is the transitive reduction that preserves reachability using only cover relations.

The Hasse dataset contains only the cover relations, i.e., the edges of the Hasse diagram of $(Y_x, \succ_x)$. We call it a skeleton because every non-cover preference in $\mathcal{D}_{\text{full}}^x$ can be recovered through a transitive path of cover relations. In this precise sense, it is the minimal skeleton of the prompt-wise order. Hasse pairs are therefore local comparisons and, in our tiered construction, correspond to small-gap pairs. However, they are not selected merely by a small-gap heuristic. Their selection criterion is structural: they are exactly the cover relations in the transitive reduction of the prompt-wise partial order. Thus, their intended role is not only to mine difficult examples, but to provide a minimal reachability-preserving anchor for local preference

consistency after broader coverage has been learned. In the special case where the responses for a prompt x form a total order

$$y_1 \succ_x y_2 \succ_x \cdots \succ_x y_K,$$

the Hasse pairwise dataset reduces to

$$\mathcal{D}_{\text{Hasse}}^x = \{(x, y_k, y_{k+1}) : k = 1, \dots, K-1\}.$$

This transitivity-induced structure can be used as an inductive bias for learning. Since the comparable-pair order is determined by the Hasse skeleton, it is sufficient, under the transitive prompt-wise relation, to correctly rank the Hasse pairs rather than every pair in the full dataset. Thus, in the total-order case, the number of comparisons per prompt is reduced from $O(K^2)$ to $O(K)$ while preserving the information needed to recover $\mathcal{D}_{\text{full}}^x$ under transitivity. The next proposition formalizes this control of comparable-pair error by Hasse-ranking error.

Remark. This is a within-prompt structural statement, not a claim that training only on Hasse pairs is sufficient for best held-out generalization. Hasse pairs are local and typically hard; in Section 5.1, Hasse-only continuation is therefore treated as an anchor or refinement stage, and is compared against coverage-preserving and auxiliary alternatives.

Proposition 4.1 (Hasse ranking controls comparable-pair error). *Fix a prompt x and assume that $(Y_x, \succ_x)$ is finite and transitive. Then, for every scoring model θ ,*

$$\mathcal{R}_{\text{full}}^x(\theta) \leq |\mathcal{D}_{\text{Hasse}}^x| \mathcal{R}_{\text{Hasse}}^x(\theta).$$

Consequently, if $\mathcal{R}_{\text{Hasse}}^x(\theta_t) \rightarrow 0$, then $\mathcal{R}_{\text{full}}^x(\theta_t) \rightarrow 0$.

Proof. The proof is given in Appendix A. □

Proposition 4.2 (Hasse-containing loss controls Hasse ranking error). *Fix a prompt x and assume that $(Y_x, \succ_x)$ is finite and transitive. Let $\mathcal{S}^x$ be any pairwise dataset satisfying $\mathcal{D}_{\text{Hasse}}^x \subseteq \mathcal{S}^x \subseteq \mathcal{D}_{\text{full}}^x$. Define*

$$\mathcal{L}_{\mathcal{S}}^x(\theta) := \frac{1}{|\mathcal{S}^x|} \sum_{(x, y_i, y_j) \in \mathcal{S}^x} \mathcal{L}_{ij}(\theta).$$

If $\mathcal{L}_{\mathcal{S}}^x(\theta_t) \rightarrow 0$, then $\mathcal{R}_{\text{Hasse}}^x(\theta_t) \rightarrow 0$ and, by Proposition 4.1, $\mathcal{R}_{\text{full}}^x(\theta_t) \rightarrow 0$.

Proof. The proof is given in Appendix A. □

Proposition 4.3 (Minimality of the Hasse dataset). *Fix a prompt x and assume that $(Y_x, \succ_x)$ is finite and transitive. Assume the scoring model class can realize arbitrary finite prompt-wise score assignments for this fixed x . If a pairwise dataset $\mathcal{S}^x \subsetneq \mathcal{D}_{\text{full}}^x$ omits at least one Hasse edge, then there exists a sequence $\{\theta_t\}_{t=1}^{\infty}$ such that $\mathcal{L}_{\mathcal{S}}^x(\theta_t) \rightarrow 0$ but $\mathcal{R}_{\text{full}}^x(\theta_t) \not\rightarrow 0$. Consequently, any pairwise dataset whose correct ranking universally guarantees correct ranking over all comparable pairs must contain $\mathcal{D}_{\text{Hasse}}^x$.*

Proof. The proof is given in Appendix A. □

The Hasse dataset is therefore not merely a computationally cheaper subset of pairwise comparisons. Under the finite transitive prompt-wise assumptions above, it is the minimal cover-constraint skeleton for controlling errors over all comparable pairs: including all Hasse edges is sufficient by Propositions 4.1 and 4.2, and omitting any Hasse edge destroys the universal within-prompt guarantee by Proposition 4.3. In actual training, we use this skeleton as an anchor. The Hasse-containing training set is the object with direct transitive-control interpretation; auxiliary pairs are added to improve held-out performance while retaining that anchor.

4.3 Pair Types and Distributional Bias

The pair selectors used in this paper differ not only in size, but also in how they reshape the original pair distribution. Rnd is the coverage-preserving baseline. Ext and Hasse represent opposite static biases: Ext keeps large-gap pairs that should be rank-distance-easy, whereas Hasse keeps local cover relations that are expected to be small-gap and therefore harder. This is why Ext can be used before Hasse as an easy warm-start stage: it provides coarse high-versus-low separation without depending on the current model, while Rnd supplies broad distributional coverage.

Concretely, fix a prompt x and let $\pi_x : Y_x \rightarrow \{1, \dots, K\}$ be a fixed ranking or a fixed linear extension of the preference relation, where smaller values mean more preferred responses. For partial orders with incomparable responses, this linear extension is fixed before training and is used only to define a static large-gap selector. It is part of the dataset construction and does not depend on the current scoring model θ . For $k \in \{1, \dots, \lfloor K/2 \rfloor\}$, define

$$\mathcal{E}_{\text{top},k}^x := \{y_i \in Y_x : \pi_x(y_i) \leq k\}, \quad \mathcal{E}_{\text{bottom},k}^x := \{y_i \in Y_x : \pi_x(y_i) > K - k\}.$$

The Ext set is the static, model-independent set of comparable large-gap pairs between these two extremes:

$$\mathcal{D}_{\text{Ext},k}^x := \{(x, y_i, y_j) \in \mathcal{D}_{\text{full}}^x : y_i \in \mathcal{E}_{\text{top},k}^x, y_j \in \mathcal{E}_{\text{bottom},k}^x\}.$$

4.4 Auxiliary Pair Selection

The full curriculum uses a Hasse+Aux final stage, where the Hasse anchor is retained and a budget-matched non-Hasse auxiliary set is added. The Hasse-only stage is the corresponding base case used to isolate the anchor. The non-Hasse candidate pool is

$$\mathcal{C}^x := \mathcal{D}_{\text{full}}^x \setminus \mathcal{D}_{\text{Hasse}}^x.$$

Let b_x denote the prompt-wise auxiliary budget, i.e., the number of non-Hasse auxiliary pairs selected for prompt x under the matched-budget protocol described in Section 5. In the algorithm below, the same notation is also used for the matched stage budget when sampling Rnd stages; the relevant budget is always fixed by the condition being compared. All auxiliary rules select up to b_x non-Hasse pairs per prompt, using the same budget within each matched comparison. Thus, for each auxiliary rule q ,

$$\mathcal{A}_q^x \subseteq \mathcal{C}^x, \quad |\mathcal{A}_q^x| \leq b_x,$$

and the reported experiments sample all compared auxiliary conditions to the same realized prompt-wise size. We write $\text{SAMPLE}(B, b)$ for a budget- b random subset of B , and $\text{SMALLEST}(B, b; g)$ for the b elements of B with the smallest values of the score g . The two model-independent auxiliary choices used throughout the main experiments are Rnd and Ext. Rnd is a budget-matched random subset, while Ext takes the budget-matched large-gap subset from the non-Hasse pool:

$$\mathcal{A}_{\text{Rnd}}^x := \text{SAMPLE}(\mathcal{C}^x, b_x), \quad \mathcal{A}_{\text{Ext},k}^x := \text{SAMPLE}(\mathcal{D}_{\text{Ext},k}^x \cap \mathcal{C}^x, b_x).$$

We also consider two model-dependent hard auxiliary rules computed after a warm-start model θ_0 has scored the non-Hasse candidate pool. Unc selects the b_x non-Hasse pairs with smallest absolute warm-start margin, i.e., boundary cases for the current model:

$$\mathcal{A}_{\text{Unc}}^x := \text{SMALLEST}(\mathcal{C}^x, b_x; |\Delta_{ij}(\theta_0)|).$$

Mis selects the b_x non-Hasse pairs with the most negative warm-start margin, restricted to pairs currently ranked in the wrong direction:

$$\mathcal{A}_{\text{Mis}}^x := \text{SMALLEST}(\{(x, y_i, y_j) \in \mathcal{C}^x : \Delta_{ij}(\theta_0) < 0\}, b_x; \Delta_{ij}(\theta_0)).$$

In this sense Mis is the harder model-dependent subset. These rules are auxiliary only in the sense that they are added around the Hasse anchor; they are biased selected subsets and are not intended as standalone replacement data.

For any auxiliary rule q , the Hasse-anchored training set is

$$\mathcal{S}_q^x := \mathcal{D}_{\text{Hasse}}^x \sqcup \mathcal{A}_q^x.$$

4.5 Training Algorithm

Algorithm 1 summarizes the training template. The primary full schedule is $\text{Ext} \rightarrow \text{Rnd} \rightarrow \text{Hasse} + \text{Aux}$. The Hasse-only version is the base schedule used to isolate the late-anchor effect.

Algorithm 1 Hasse-anchored curriculum training

Require: Candidate response sets $\{Y_x\}_{x \in X}$; stage sequence $(c_1, \dots, c_T)$ with $c_t \in \{\text{Ext}, \text{Rnd}, \text{Hasse}, \text{Hasse} + \text{Aux}\}$; auxiliary rule q for $\text{Hasse} + \text{Aux}$ stages; budgets $\{b_x\}_{x \in X}$; Ext parameter k .

Ensure: Trained scoring model θ_T .

```

1: for each prompt  $x \in X$  do
2:    $\mathcal{D}_{\text{full}}^x \leftarrow \text{ALLCOMPARABLEPAIRS}(Y_x)$ 
3:    $\mathcal{D}_{\text{Hasse}}^x \leftarrow \text{HASSESKELTON}(\mathcal{D}_{\text{full}}^x)$ 
4:    $\mathcal{D}_{\text{Ext},k}^x \leftarrow \text{EXTREMEPAIRS}(\mathcal{D}_{\text{full}}^x, k)$ 
5: end for
6: Initialize scoring model  $\theta_0$ .
7: for  $t = 1, \dots, T$  do
8:   for each prompt  $x \in X$  do
9:     if  $c_t = \text{Ext}$  then
10:       $\mathcal{S}_t^x \leftarrow \mathcal{D}_{\text{Ext},k}^x$ 
11:     else if  $c_t = \text{Rnd}$  then
12:       $\mathcal{S}_t^x \leftarrow \text{SAMPLE}(\mathcal{D}_{\text{full}}^x, b_x)$ 
13:     else if  $c_t = \text{Hasse}$  then
14:       $\mathcal{S}_t^x \leftarrow \mathcal{D}_{\text{Hasse}}^x$ 
15:     else if  $c_t = \text{Hasse} + \text{Aux}$  then
16:       If  $q$  is model-dependent, score non-Hasse candidates with current model  $\theta_{t-1}$ .
17:        $\mathcal{A}_q^x \leftarrow \text{SELECTAUX}(\mathcal{D}_{\text{full}}^x \setminus \mathcal{D}_{\text{Hasse}}^x, q, b_x)$ 
18:        $\mathcal{S}_t^x \leftarrow \mathcal{D}_{\text{Hasse}}^x \sqcup \mathcal{A}_q^x$ 
19:     else
20:       raise unknown stage error.
21:     end if
22:   end for
23:   Continue training on  $\bigsqcup_x \mathcal{S}_t^x$  to obtain  $\theta_t$ .
24: end for
25: return  $\theta_T$ 

```

5 Experiments

We evaluate on a controlled contextual preference task built from MovieLens 25M Tag Genome data (Harper & Konstan, 2015; Vig et al., 2012). Each movie is a context x , each tag is an item y , and the Tag Genome relevance score is discretized into 10 preference tiers. For each data seed, we use 50 training movies and 20 disjoint held-out movies; each movie contains 100 tags balanced across tiers. Cross-tier tag pairs induce preferences, while same-tier tags are not compared. This yields 4,500 full pairs and 900 adjacent-tier Hasse pairs per movie. Full accuracy averages over all held-out cross-tier pairs, while Gap1 accuracy evaluates only adjacent-tier held-out pairs and is used as the diagnostic for local ranking quality. We also report standard rank-correlation and top-ranking metrics, including Spearman, Kendall, and NDCG (Spearman, 1904; Kendall, 1938; Järvelin & Kekäläinen, 2002). All matched comparisons use the same prompt-wise pair budget, so differences between conditions reflect selector or anchor choice rather than the number of final-stage pairs.

We use RoBERTa-base (Liu et al., 2019) with a scalar scoring head, trained with the pairwise logistic loss from Definition 3.2; implementation details are given in Appendix B. We report means with standard errors over matched runs and use paired tests for significance claims. Full trajectories are shown only for diagnostics; otherwise, post-switch plots focus on the final continuation stage. Because the realized pair

counts, and therefore the number of optimizer steps, can vary across data seeds, trajectory figures align intermediate checkpoints on the logged step grid and mark final evaluations separately.

5.1 Hasse as Small-Gap Supervision

In this section, we characterize Hasse before using it inside the full curriculum. We ask the same question at different stages of training: what happens when the model is trained on the local Hasse skeleton? From scratch, Hasse behaves like hard small-gap supervision and is much worse than broad random data. After a random warm-up, however, the same Hasse pairs become useful as a local continuation set. This contrast motivates treating Hasse as a hard curriculum stage rather than as a general-purpose replacement for random supervision.

5.1.1 Hasse from Scratch

We test whether Hasse pairs are useful without curriculum. We train from the pretrained model for one epoch on either the full Hasse set or a movie-level budget-matched random set. This diagnostic isolates the small-gap nature of Hasse supervision before any broad warm-up has been learned.

Table 1: No-curriculum diagnostic. Both rows train for one epoch from the pretrained model with the same movie-level pair budget. Values are mean $\pm$ standard error over 16 matched runs.

Method	Full	Gap1	Spearman	Kendall	NDCG@10	NDCG@20
Rnd	0.7970 ± 0.0010	0.6169 ± 0.0009	0.7349 ± 0.0025	0.5493 ± 0.0021	0.5423 ± 0.0104	0.6177 ± 0.0075
Hasse	0.5533 ± 0.0085	0.5162 ± 0.0031	0.1456 ± 0.0232	0.0986 ± 0.0158	0.2349 ± 0.0136	0.2917 ± 0.0143

Hasse is substantially worse than budget-matched random training from scratch: the paired differences Hasse–Rnd are negative on every metric and every matched run, including full accuracy (-0.2437 , $p < 10^{-13}$), Gap1 (-0.1007 , $p < 10^{-14}$), Spearman (-0.5893 , $p < 10^{-13}$), Kendall (-0.4507 , $p < 10^{-13}$), and NDCG@20 (-0.3260 , $p < 10^{-12}$). This confirms that Hasse is not a standalone replacement for broad random supervision. Used too early, it behaves as a difficult small-gap training set rather than as a general-purpose data source. Thus, although the Hasse skeleton gives the structural transitive-control guarantee developed in Section 4, it is still hard local supervision in optimization terms. Curriculum design should therefore treat Hasse like a hard-sample stage: it is unlikely to help early training before the model has learned broad coverage.

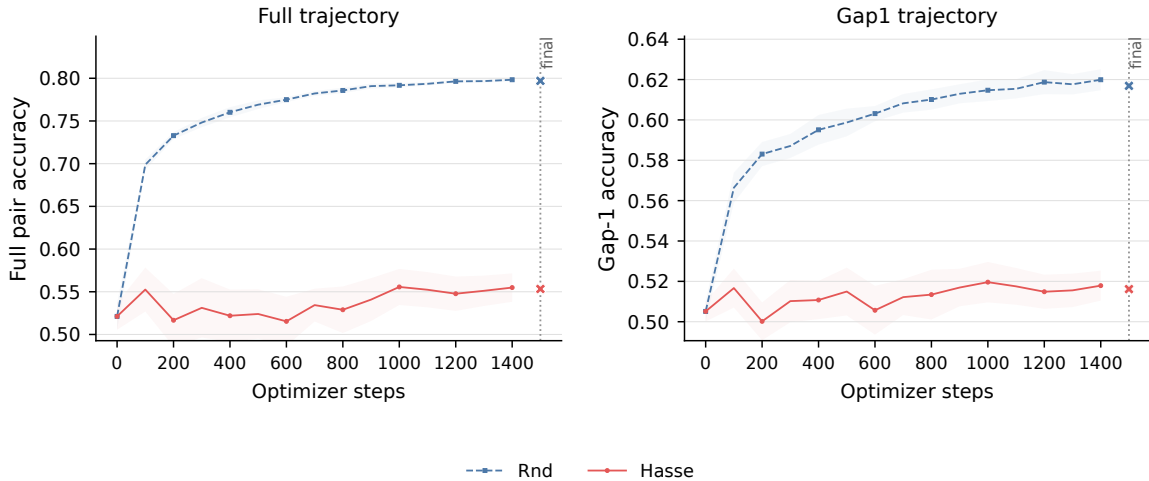


Figure 2: Held-out accuracy trajectories without curriculum.

5.1.2 Hasse after Warm-Up

We isolate the role of Hasse without the additional effects of Ext or auxiliary data. Both conditions first receive the same random warm-up epoch, which gives the model broad coverage of the pair distribution. We then compare two budget-matched continuations: another random epoch, or an epoch on the Hasse cover pairs. In contrast to Section 5.1.1, this evaluates Hasse after the model has already learned broad coverage from random pairs. If Hasse is a hard local supervision set whose usefulness depends on curriculum placement, the Hasse continuation should improve local ranking metrics without relying on a larger training budget.

Table 2: Anchor experiment after one random warm-up epoch. Values are mean $\pm$ standard error over 16 matched runs.

Method	Full	Gap1	Spearman	Kendall	NDCG@10	NDCG@20
Rnd→Hasse	0.8042 ± 0.0012	0.6243 ± 0.0010	0.7464 ± 0.0028	0.5627 ± 0.0025	0.5747 ± 0.0109	0.6429 ± 0.0075
Rnd→Rnd	0.8021 ± 0.0014	0.6205 ± 0.0011	0.7434 ± 0.0031	0.5585 ± 0.0028	0.5714 ± 0.0096	0.6356 ± 0.0068

Hasse continuation improves over a budget-matched random continuation on most metrics, with small but consistent gains in full accuracy ($+0.0020$, $p = 0.0002$), Gap1 ($+0.0038$, $p = 0.00002$), Spearman ($+0.0030$, $p = 0.0008$), Kendall ($+0.0042$, $p = 0.00005$), and NDCG@20 ($+0.0073$, $p = 0.018$). Together with Section 5.1.1, this shows that Hasse supervision is stage-dependent: it is too narrow and difficult to drive early learning by itself, but after broad warm-up it can refine local preference consistency. We therefore next ask whether Hasse is most useful when preceded by coarse and coverage-oriented stages, and later whether the final Hasse stage benefits from auxiliary non-Hasse supervision.

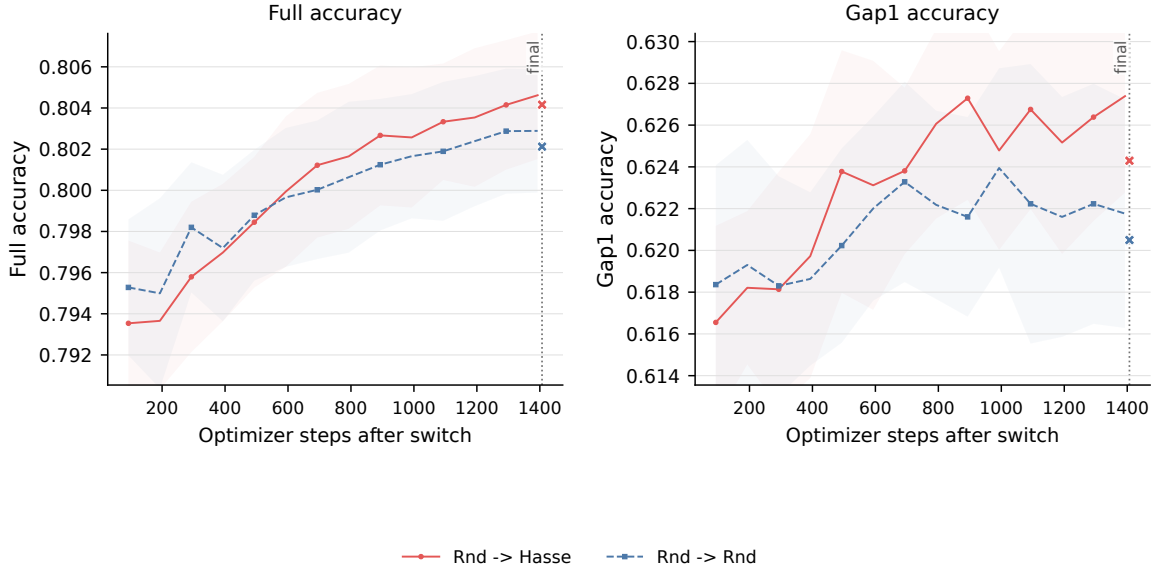


Figure 3: Post-switch accuracy trajectories for Hasse continuation after random warm-up.

5.2 Base Curriculum Ordering

In this section, we evaluate the base ordering before adding auxiliary data to the final Hasse stage. This follows the standard curriculum-learning intuition that easier examples can be useful before harder examples (Bengio et al., 2009; Kumar et al., 2010; Jiang et al., 2015). The hypothesis is that pair types are most useful when introduced in the order suggested by their roles: Ext for easy coarse separation, Rnd for broad

coverage, and Hasse for late local refinement. We compare this Hasse-only base curriculum against matched alternatives that replace either the early Ext stage or the late Hasse stage with random data.

Table 3: Curriculum experiment comparing early Ext and late Hasse. Values are mean $\pm$ standard error over 16 matched runs.

Method	Full	Gap1	Spearman	Kendall	NDCG@10	NDCG@20
Ext $\rightarrow$ Rnd $\rightarrow$ Hasse	0.8051 $\pm$ 0.0009	0.6261 $\pm$ 0.0011	0.7473 $\pm$ 0.0022	0.5644 $\pm$ 0.0019	0.5798 $\pm$ 0.0110	0.6440 $\pm$ 0.0079
Ext $\rightarrow$ Rnd $\rightarrow$ Rnd	0.8031 $\pm$ 0.0009	0.6230 $\pm$ 0.0007	0.7440 $\pm$ 0.0022	0.5603 $\pm$ 0.0019	0.5688 $\pm$ 0.0101	0.6345 $\pm$ 0.0071
Rnd $\rightarrow$ Rnd $\rightarrow$ Hasse	0.8046 $\pm$ 0.0008	0.6251 $\pm$ 0.0012	0.7467 $\pm$ 0.0020	0.5635 $\pm$ 0.0018	0.5691 $\pm$ 0.0104	0.6369 $\pm$ 0.0081
Rnd $\rightarrow$ Rnd $\rightarrow$ Rnd	0.8025 $\pm$ 0.0010	0.6212 $\pm$ 0.0010	0.7431 $\pm$ 0.0023	0.5591 $\pm$ 0.0020	0.5622 $\pm$ 0.0102	0.6294 $\pm$ 0.0072

Ext $\rightarrow$ Rnd $\rightarrow$ Hasse is the strongest average condition across all reported metrics. Conditional on Ext and Rnd already being used, replacing the final random epoch with Hasse improves full accuracy by 0.0021 ($p = 0.0005$), Gap1 by 0.0031 ($p = 0.0029$), Spearman by 0.0032 ($p = 0.0012$), Kendall by 0.0041 ($p = 0.0002$), NDCG@10 by 0.0110 ($p = 0.025$), and NDCG@20 by 0.0095 ($p = 0.0039$). The same late-Hasse effect appears after a random warm-up: Rnd $\rightarrow$ Rnd $\rightarrow$ Hasse improves over Rnd $\rightarrow$ Rnd $\rightarrow$ Rnd on every metric, including Gap1 (+0.0039, $p = 0.00002$). We interpret the Hasse stage as late local refinement and top-ranking stabilization rather than broad distribution learning by itself.

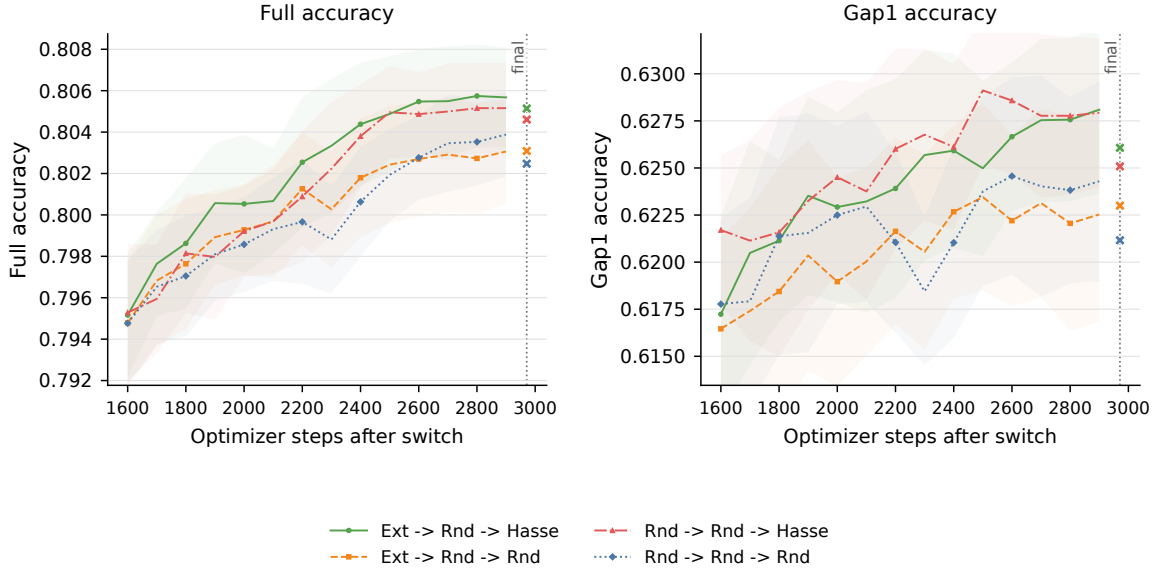


Figure 4: Post-switch accuracy trajectories for the base curriculum ordering experiment.

5.3 Why the Hasse Anchor is Needed

In this section, we directly test the central claim that late-stage auxiliary selection requires an anchor. We first ask whether a selected non-Hasse auxiliary subset can be used by itself after the Ext $\rightarrow$ Rnd warm-up. We then compare Hasse+Aux against Rnd+Aux while keeping the auxiliary distribution fixed. This separates two questions: whether Aux-only continuation is stable, and whether Hasse is a better final-stage backbone than a budget-matched random set.

5.3.1 Aux-Only Continuation

The first diagnostic removes both Hasse and random backbone pairs from the final stage and trains only on the selected auxiliary subset: Ext→Rnd→Aux. This is not intended as a budget-matched replacement for Hasse+Aux; instead, it asks whether the auxiliary subset alone is sufficient to support late-stage continuation. If auxiliary selectors reshape the pair distribution too aggressively, the final stage may become unstable without a separate anchor that preserves broader order structure.

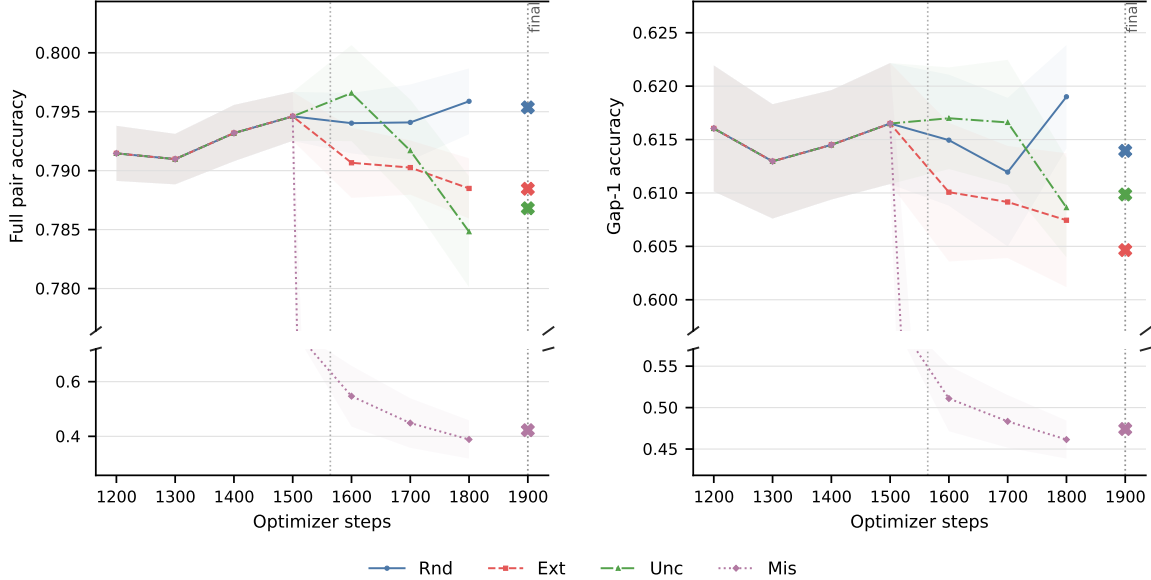


Figure 5: Late-stage Aux-only trajectories after the shared Ext→Rnd warm-up. The broken axis exposes the collapse of Mis-only continuation while preserving the scale needed to compare Rnd, Ext, and Unc.

Figure 5 shows that Aux-only continuation is not a stable explanation for the final-stage gains. Mis-only training collapses after the switch, ending at full accuracy 0.423 and Gap1 accuracy 0.474. The other auxiliary choices are less extreme, but they do not produce a clear stable improvement during the final stage. The Mis case is especially informative: when the late-stage data are selected from a strongly biased subset, continued training can move the model away from the broader preference distribution rather than refine it. This supports the view that selected auxiliary data can narrow or distort the late training distribution, and should therefore be paired with an anchor or backbone that keeps broader order structure present. The natural next question is what kind of anchor should play this role. The Hasse dataset is a plausible candidate because it retains the adjacent preference relations needed to preserve prompt-wise order structure while remaining much smaller than the full transitive closure.

5.3.2 Hasse versus Other Anchors

We next test whether the Hasse set itself is a useful anchor, rather than only a proxy for generic small-gap training. In our tiered construction, the Hasse set contains all gap-1 comparisons for each prompt. We therefore ask whether the late-stage gains come from using this complete Hasse anchor, or merely from exposing the model to some small-gap pairs. All anchor budgets are matched at the prompt level, and each paired comparison keeps the same auxiliary selector while changing only the anchor component of the final stage.

We compare Hasse with four control anchors. Rnd samples budget-matched pairs from the full pair pool, so it may include some Hasse comparisons but does not guarantee the complete Hasse anchor. SmallGap samples from gap-1–3 pairs and is the closest small-gap control, since it can include Hasse comparisons while

also replacing some of them with wider local pairs. NonHasseRnd and NonHasseSmallGap are negative controls that remove Hasse comparisons whenever possible. These controls separate the effect of general random coverage, local small-gap supervision, and explicit retention of the complete Hasse anchor.

Table 4: Final-stage anchor comparison. Each entry is the paired difference $Hasse+Aux$ minus $Control+Aux$ under the same auxiliary selector, after the shared Ext→Rnd warm-up. Positive values favor Hasse.

Control anchor	Aux	Full	Gap1	Spearman	Kendall	NDCG@10	NDCG@20
NonHasseRnd	Unc	+0.0025±0.0008	+0.0057±0.0009	+0.0030±0.0016	+0.0053±0.0015	+0.0298±0.0061	+0.0230±0.0050
	Mis	+0.0040±0.0009	+0.0050±0.0010	+0.0068±0.0018	+0.0082±0.0017	+0.0227±0.0080	+0.0173±0.0050
	Rnd	+0.0046±0.0006	+0.0056±0.0009	+0.0073±0.0013	+0.0088±0.0011	+0.0264±0.0073	+0.0174±0.0059
	Ext	+0.0042±0.0008	+0.0070±0.0011	+0.0059±0.0016	+0.0083±0.0014	+0.0275±0.0063	+0.0204±0.0047
NonHasseSmallGap	Unc	+0.0007±0.0008	+0.0032±0.0011	+0.0001±0.0015	+0.0019±0.0015	+0.0329±0.0087	+0.0205±0.0063
	Mis	+0.0008±0.0008	+0.0025±0.0010	+0.0007±0.0015	+0.0020±0.0015	+0.0266±0.0063	+0.0191±0.0039
	Rnd	+0.0038±0.0009	+0.0062±0.0009	+0.0056±0.0018	+0.0076±0.0018	+0.0241±0.0073	+0.0168±0.0058
	Ext	+0.0041±0.0009	+0.0060±0.0008	+0.0062±0.0019	+0.0084±0.0017	+0.0288±0.0076	+0.0189±0.0055
Rnd	Unc	+0.0007±0.0007	+0.0019±0.0009	+0.0007±0.0015	+0.0021±0.0014	+0.0125±0.0049	+0.0099±0.0046
	Mis	+0.0023±0.0006	+0.0035±0.0009	+0.0038±0.0012	+0.0047±0.0011	+0.0141±0.0052	+0.0086±0.0035
	Rnd	+0.0027±0.0009	+0.0035±0.0011	+0.0044±0.0017	+0.0051±0.0018	+0.0130±0.0070	+0.0090±0.0049
	Ext	+0.0025±0.0010	+0.0038±0.0013	+0.0037±0.0017	+0.0051±0.0019	+0.0309±0.0124	+0.0200±0.0086
SmallGap	Unc	-0.0011±0.0006	+0.0005±0.0006	-0.0029±0.0011	-0.0015±0.0011	+0.0107±0.0061	+0.0068±0.0050
	Mis	+0.0001±0.0005	+0.0011±0.0008	-0.0001±0.0010	+0.0004±0.0011	+0.0122±0.0051	+0.0032±0.0038
	Rnd	+0.0002±0.0008	+0.0013±0.0008	-0.0004±0.0016	+0.0003±0.0015	+0.0063±0.0064	+0.0043±0.0058
	Ext	-0.0003±0.0007	+0.0012±0.0007	-0.0012±0.0014	-0.0001±0.0012	+0.0108±0.0063	+0.0061±0.0043

Table 5: Effect of allowing Hasse comparisons inside a random anchor. Each entry is $Rnd+Aux$ minus $NonHasseRnd+Aux$.

Aux	Full	Gap1	Spearman	Kendall	NDCG@10	NDCG@20
Unc	+0.0019±0.0006	+0.0038±0.0008	+0.0023±0.0014	+0.0032±0.0013	+0.0173±0.0060	+0.0131±0.0045
Mis	+0.0018±0.0007	+0.0015±0.0010	+0.0030±0.0015	+0.0034±0.0014	+0.0087±0.0076	+0.0087±0.0045
Rnd	+0.0019±0.0007	+0.0021±0.0009	+0.0029±0.0014	+0.0038±0.0014	+0.0135±0.0054	+0.0084±0.0047
Ext	+0.0017±0.0010	+0.0032±0.0013	+0.0022±0.0020	+0.0032±0.0020	-0.0034±0.0121	+0.0004±0.0082

Table 6: Effect of allowing Hasse comparisons inside a small-gap anchor. Each entry is $SmallGap+Aux$ minus $NonHasseSmallGap+Aux$.

Aux	Full	Gap1	Spearman	Kendall	NDCG@10	NDCG@20
Unc	+0.0018±0.0007	+0.0027±0.0010	+0.0030±0.0012	+0.0034±0.0013	+0.0222±0.0080	+0.0137±0.0040
Mis	+0.0007±0.0008	+0.0015±0.0007	+0.0008±0.0016	+0.0016±0.0015	+0.0144±0.0053	+0.0159±0.0034
Rnd	+0.0036±0.0008	+0.0048±0.0008	+0.0060±0.0016	+0.0073±0.0015	+0.0177±0.0055	+0.0125±0.0035
Ext	+0.0044±0.0009	+0.0048±0.0008	+0.0074±0.0017	+0.0084±0.0016	+0.0180±0.0075	+0.0128±0.0050

Tables 4–6 show that removing gap-1 Hasse comparisons from the anchor pool consistently hurts performance. The largest drops occur for NonHasseRnd and NonHasseSmallGap, where the final-stage anchor no longer preserves Hasse structure. Tables 5 and 6 make the same effect visible without using Hasse as the left-hand method: allowing Hasse comparisons into either a random or small-gap anchor improves most metrics, especially Gap1. Hasse also improves over Rnd for most metrics, indicating that random budget matching

is not enough. The strictest comparison is SmallGap: it contains many local pairs and can include Hasse comparisons, but it does not force the anchor to contain the complete Hasse anchor. The smaller but mostly positive differences against SmallGap suggest that Hasse is not just a convenient name for small-gap learning; its usefulness comes from preserving the complete adjacent-preference structure as a stable late-stage anchor.

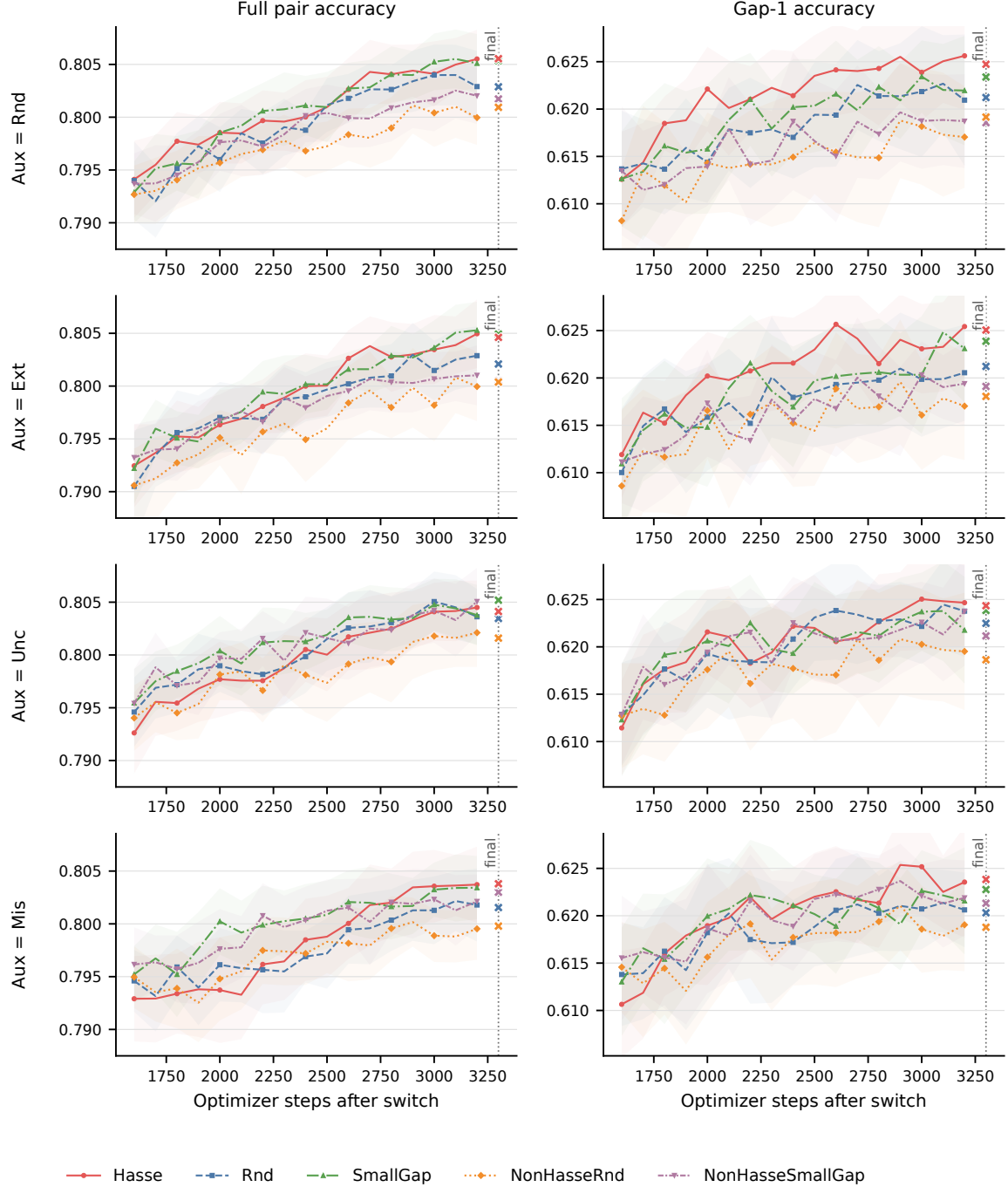


Figure 6: Post-switch accuracy trajectories for the final-stage anchor comparison. Each panel compares Hasse+Aux with budget-matched control-anchor variants under the same auxiliary selector.

5.4 Full Curriculum with Auxiliary Hasse Anchor

Having established the need for an anchor in Section 5.3, we ask which auxiliary distribution is most useful around the Hasse anchor. This section combines the strongest ordering from Section 5.2 with Hasse+Aux final-stage variants. All conditions use the same Ext→Rnd warm-up. The final stage then uses either Hasse+Rnd, Hasse+Ext, Hasse+Unc, or Hasse+Mis. Because the Mis pool size depends on the warm-start model, the auxiliary budget is matched separately within each data/training seed; all auxiliary choices in a matched run therefore use the same final-stage budget.

Table 7: Final curriculum auxiliary experiment. All rows use Ext→Rnd warm-up followed by the listed final-stage data. Values are mean $\pm$ standard error over 16 matched runs.

Method	Full	Gap1	Spearman	Kendall	NDCG@10	NDCG@20
Ext→Rnd→Hasse+Rnd	0.8055 $\pm$ 0.0011	0.6247 $\pm$ 0.0010	0.7489 $\pm$ 0.0026	0.5647 $\pm$ 0.0024	0.5738 $\pm$ 0.0113	0.6399 $\pm$ 0.0073
Ext→Rnd→Hasse+Ext	0.8046 $\pm$ 0.0013	0.6251 $\pm$ 0.0010	0.7465 $\pm$ 0.0031	0.5635 $\pm$ 0.0026	0.5772 $\pm$ 0.0119	0.6416 $\pm$ 0.0082
Ext→Rnd→Hasse+Unc	0.8041 $\pm$ 0.0012	0.6243 $\pm$ 0.0009	0.7457 $\pm$ 0.0029	0.5624 $\pm$ 0.0025	0.5820 $\pm$ 0.0110	0.6434 $\pm$ 0.0080
Ext→Rnd→Hasse+Mis	0.8038 $\pm$ 0.0013	0.6238 $\pm$ 0.0010	0.7454 $\pm$ 0.0032	0.5618 $\pm$ 0.0027	0.5774 $\pm$ 0.0102	0.6396 $\pm$ 0.0071

The final auxiliary experiment shows how the choice of auxiliary data changes the Hasse-anchored final stage. Hasse+Rnd gives the best full accuracy, Spearman, and Kendall; Hasse+Ext gives the best Gap1; and Hasse+Unc gives the best NDCG scores. Hasse+Mis is competitive but does not dominate any metric. Thus, after the Ext→Rnd warm-up, the Hasse anchor is robust across auxiliary choices, while the auxiliary set determines whether the final stage emphasizes coverage, easy separation, boundary cases, or model-dependent errors.

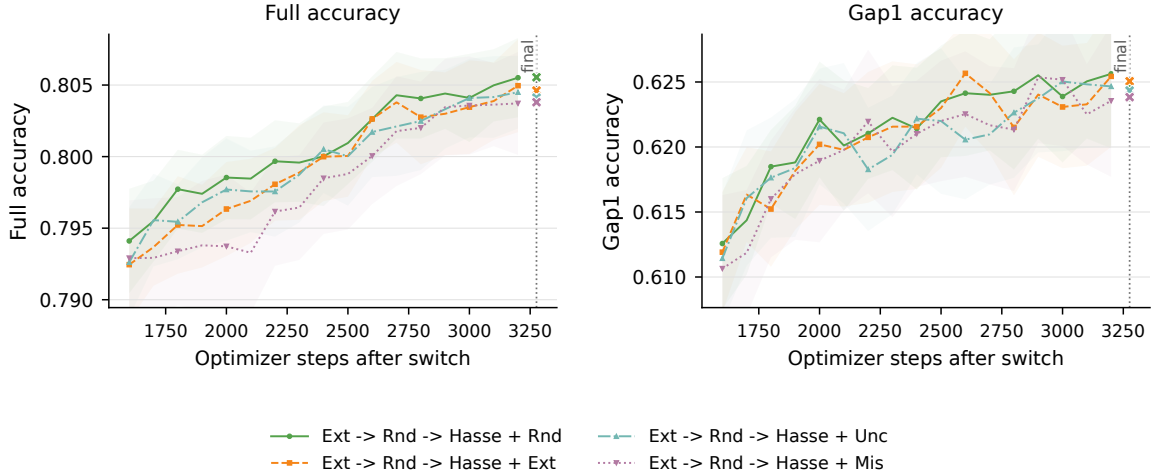


Figure 7: Post-switch accuracy trajectories for the full Hasse+Aux curriculum.

6 Conclusion

We presented a Hasse-anchored curriculum for controlled pairwise preference learning. The theory identifies Hasse pairs as the minimal transitive skeleton that must be present for direct control over all comparable pairs within a finite prompt-wise preference relation, while the experiments show when this structure is most useful in practice. Hasse continuation improves over random continuation after warm-up, and its strongest role is as a late local anchor. The strongest practical curriculum first uses Ext for coarse separation, then Rnd for broad coverage, and finally Hasse+Aux for local refinement with complementary non-Hasse supervision.

6.1 Discussion and Limitations

Across these experiment groups, the same pattern appears in the controlled MovieLens preference benchmark. The strongest practical curriculum is staged: use Ext for coarse early separation, use random data to preserve broad coverage, and introduce a Hasse+Aux final stage to stabilize adjacent and top-ranking structure without relying on Hasse alone. Hasse-only continuation is useful as a diagnostic base case after warm-up, but its role is best understood as local anchoring rather than broad distribution construction. Unc and Mis identify difficult regions of the current model, but concentrating on those regions too early can distort the training distribution and hurt held-out ranking.

The main limitation is that the benchmark is deliberately controlled: the preference relation is finite, prompt-wise transitive, and induced by tiered MovieLens relevance scores. This design lets us isolate the role of Hasse structure, but it does not by itself establish the same behavior for noisy, cyclic, or annotator-dependent human preferences. Similarly, reward modeling and preference optimization for language models are natural downstream settings for this curriculum view, but the present experiments should be read as evidence about controlled pairwise preference learning rather than as a direct downstream RLHF claim. Extending Hasse-anchored curricula to noisy human-feedback data is an important next step.

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A Deferred Proofs

Proposition 3.1 (restated). For any θ ,

$$\mathcal{R}(\theta) \leq \frac{1}{\log 2} \mathcal{L}_{\text{rank}}(\theta).$$

Consequently, if $\mathcal{L}_{\text{rank}}(\theta_t) \rightarrow 0$ for a sequence $\{\theta_t\}_{t=1}^\infty$, then $\mathcal{R}(\theta_t) \rightarrow 0$.

Proof of Proposition 3.1. If $\Delta_{ij}(\theta) < 0$, then $\sigma(\Delta_{ij}(\theta)) < 1/2$ and hence

$$\mathbf{1}[\Delta_{ij}(\theta) < 0] \leq -\frac{\log \sigma(\Delta_{ij}(\theta))}{\log 2}.$$

This gives the pointwise inequality

$$\mathbf{1}[\Delta_{ij}(\theta) < 0] \leq \frac{1}{\log 2} \mathcal{L}_{ij}(\theta).$$

Taking expectations over $\mathcal{D}$ gives

$$\begin{aligned} \mathcal{R}(\theta) &= \mathbb{E}_{\mathcal{D}} [\mathbf{1}[\Delta_{ij}(\theta) < 0]] \\ &\leq \frac{1}{\log 2} \mathbb{E}_{\mathcal{D}} [\mathcal{L}_{ij}(\theta)] = \frac{1}{\log 2} \mathcal{L}_{\text{rank}}(\theta). \end{aligned}$$

Therefore, $\mathcal{L}_{\text{rank}}(\theta_t) \rightarrow 0$ implies $\mathcal{R}(\theta_t) \rightarrow 0$ by the squeeze theorem. $\square$

Proposition 4.1 (restated). Fix a prompt x and assume that $(Y_x, \succ_x)$ is finite and transitive. Then, for every scoring model θ ,

$$\mathcal{R}_{\text{full}}^x(\theta) \leq |\mathcal{D}_{\text{Hasse}}^x| \mathcal{R}_{\text{Hasse}}^x(\theta).$$

Consequently, if $\mathcal{R}_{\text{Hasse}}^x(\theta_t) \rightarrow 0$, then $\mathcal{R}_{\text{full}}^x(\theta_t) \rightarrow 0$.

Proof of Proposition 4.1. For any $(x, y_i, y_j) \in \mathcal{D}_{\text{full}}^x$, finiteness and transitivity imply that there exists a cover chain

$$y_i = z_0 \succ_x z_1 \succ_x \cdots \succ_x z_q = y_j.$$

By telescoping,

$$D_i - D_j = \sum_{r=0}^{q-1} (D_\theta(x, z_r) - D_\theta(x, z_{r+1})).$$

Hence, if $D_i - D_j < 0$, then at least one Hasse edge in the chain must satisfy $D_\theta(x, z_r) - D_\theta(x, z_{r+1}) < 0$. Therefore,

$$\mathbf{1}[D_i - D_j < 0] \leq \sum_{(x, y_a, y_b) \in \mathcal{D}_{\text{Hasse}}^x} \mathbf{1}[D_a - D_b < 0].$$

Averaging over $\mathcal{D}_{\text{full}}^x$ gives

$$\mathcal{R}_{\text{full}}^x(\theta) \leq |\mathcal{D}_{\text{Hasse}}^x| \mathcal{R}_{\text{Hasse}}^x(\theta).$$

This proves the claimed bound, and the convergence implication follows immediately. $\square$

Proposition 4.2 (restated). Fix a prompt x and assume that $(Y_x, \succ_x)$ is finite and transitive. Let $\mathcal{S}^x$ be any pairwise dataset satisfying $\mathcal{D}_{\text{Hasse}}^x \subseteq \mathcal{S}^x \subseteq \mathcal{D}_{\text{full}}^x$. Define

$$\mathcal{L}_{\mathcal{S}}^x(\theta) := \frac{1}{|\mathcal{S}^x|} \sum_{(x, y_i, y_j) \in \mathcal{S}^x} \mathcal{L}_{ij}(\theta).$$

If $\mathcal{L}_{\mathcal{S}}^x(\theta_t) \rightarrow 0$, then $\mathcal{R}_{\text{Hasse}}^x(\theta_t) \rightarrow 0$ and, by Proposition 4.1, $\mathcal{R}_{\text{full}}^x(\theta_t) \rightarrow 0$.

Proof of Proposition 4.2. Define

$$\mathcal{R}_{\mathcal{S}}^x(\theta) := \frac{1}{|\mathcal{S}^x|} \sum_{(x, y_i, y_j) \in \mathcal{S}^x} \mathbf{1}[D_i - D_j < 0].$$

Since $\mathcal{D}_{\text{Hasse}}^x \subseteq \mathcal{S}^x$,

$$\mathcal{R}_{\text{Hasse}}^x(\theta) \leq \frac{|\mathcal{S}^x|}{|\mathcal{D}_{\text{Hasse}}^x|} \mathcal{R}_{\mathcal{S}}^x(\theta).$$

For every selected pair, if $D_i - D_j < 0$, then $\mathcal{L}_{ij}(\theta) \geq \log 2$, so we have the pointwise bound

$$\mathbf{1}[D_i - D_j < 0] \leq \frac{1}{\log 2} \mathcal{L}_{ij}(\theta).$$

Averaging over $\mathcal{S}^x$ yields $\mathcal{R}_{\mathcal{S}}^x(\theta) \leq (\log 2)^{-1} \mathcal{L}_{\mathcal{S}}^x(\theta)$. Therefore,

$$\mathcal{R}_{\text{Hasse}}^x(\theta) \leq \frac{|\mathcal{S}^x|}{|\mathcal{D}_{\text{Hasse}}^x| \log 2} \mathcal{L}_{\mathcal{S}}^x(\theta).$$

Thus $\mathcal{L}_{\mathcal{S}}^x(\theta_t) \rightarrow 0$ implies $\mathcal{R}_{\text{Hasse}}^x(\theta_t) \rightarrow 0$. The comparable-pair convergence follows from Proposition 4.1. $\square$

Proposition 4.3 (restated). Fix a prompt x and assume that $(Y_x, \succ_x)$ is finite and transitive. Assume the scoring model class can realize arbitrary finite prompt-wise score assignments for this fixed x . If a pairwise dataset $\mathcal{S}^x \subsetneq \mathcal{D}_{\text{full}}^x$ omits at least one Hasse edge, then there exists a sequence $\{\theta_t\}_{t=1}^{\infty}$ such that $\mathcal{L}_{\mathcal{S}}^x(\theta_t) \rightarrow 0$ but $\mathcal{R}_{\text{full}}^x(\theta_t) \not\rightarrow 0$. Consequently, any pairwise dataset whose correct ranking universally guarantees correct ranking over all comparable pairs must contain $\mathcal{D}_{\text{Hasse}}^x$.

Proof of Proposition 4.3. Suppose, for contradiction, that there exists a Hasse edge $(x, y_a, y_b) \in \mathcal{D}_{\text{Hasse}}^x$ such that $(x, y_a, y_b) \notin \mathcal{S}^x$. Since $y_a \succ_x y_b$, there is no response y_k satisfying $y_a \succ_x y_k \succ_x y_b$. Therefore, after removing the cover edge (y_a, y_b) , there is no directed path from y_a to y_b using only pairs in $\mathcal{S}^x$.

Hence, adding the reversed constraint $y_b \succ y_a$ to the directed graph defined by $\mathcal{S}^x$ does not create a cycle. Thus there exists a real-valued score function $r : Y_x \rightarrow \mathbb{R}$ such that

$$r(y_i) > r(y_j) \quad \text{for all } (x, y_i, y_j) \in \mathcal{S}^x, \quad r(y_b) > r(y_a).$$

Now choose a sequence of scoring models whose prompt-wise scores satisfy $D_{\theta_t}(x, y) = tr(y)$. Then every pair in $\mathcal{S}^x$ is ranked correctly with a margin that grows to infinity. Therefore, $\mathcal{L}_{\mathcal{S}}^x(\theta_t)$ converges to zero. However, the missing Hasse pair (x, y_a, y_b) is ranked incorrectly for all t because $D_{\theta_t}(x, y_a) - D_{\theta_t}(x, y_b) < 0$.

Thus, the ranking error on $\mathcal{S}^x$ converges to zero, but the error over all comparable pairs remains bounded below by $1/|\mathcal{D}_{\text{full}}^x|$. This gives the desired counterexample. Consequently, any dataset with a universal comparable-pair ranking guarantee cannot omit a Hasse edge. $\square$

B Experimental Details

All main experiments use RoBERTa-base (Liu et al., 2019) with a scalar scoring head and the pairwise logistic loss from Definition 3.2. We train with AdamW (Loshchilov & Hutter, 2019), learning rate 5×10^{-5} , weight decay 0.01, batch size 32, evaluation batch size 64, maximum sequence length 96, and one epoch per curriculum stage. Main experiments use data seeds {41, 42, 43, 44} and training seeds {41, 42, 43, 44}. An anonymous code release for reproducing the experiments is available at https://github.com/GGYLVLV/hasse_anchored_curriculum.